

2-Mark Questions & Answers — DBMS Module 5: Query Optimization & Transaction Concurrency Control

Q1. What is query optimization?

Query Optimization is the DBMS process of selecting the most efficient physical evaluation plan for a given SQL query to minimize Disk I/O and CPU execution time.

Q2. Define transaction and ACID properties.

A **Transaction** is a logical unit of database work.

- **ACID**: Atomicity (All or Nothing), Consistency (State Validity), Isolation (Independent Execution), Durability (Persistent Survival).

Q3. What is conflict serializability?

A schedule is **Conflict Serializable** if it is conflict equivalent to some serial schedule (it can be transformed into a serial schedule by swapping non-conflicting operations).

Q4. Define Shared and Exclusive locks.

- **Shared Lock (S)**: Allows multiple transactions to read a data item concurrently.
 - **Exclusive Lock (X)**: Grants exclusive read/write access to a single transaction; blocks all other access.
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Q5. What is the phantom phenomenon?

The **Phantom Phenomenon** occurs when a concurrent transaction inserts or deletes tuples satisfying a search condition while another transaction is performing range queries over that condition.
